

3.5 Shapes of molecules homework — answers

Section A — Multiple choice

A1 = A

A2 = D

A3 = C

A4 = B

A5 = B

A6 = C

A7 = B

A8 = D

Section B — Mark scheme clips

B1(a)(iii) [1]

1(a)(iii)	180°	1
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B2(b)(i) [2]

2(b)(i)	M1 1 shared pair of electrons between each S-Cl shown as •• M2 6 non-bonding electrons around each Cl shown as × AND 4 non-bonding electrons around S shown as •	2
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B2(b)(ii) [2]

2(b)(ii)	M1 shape non-linear M2 bond angle 103–105(°)	2
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B4(e) [2]

1(e)	M1 6 bonding pairs (of electrons and 0 lone pairs) OR coordination number = 6 OR $x = 6$ M2 bonding pairs repel (equally)	2
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B5(b)(ii) [1]

1(b)(ii)	trigonal bipyramid(al)	1
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B6(b)(i) [2]

4(b)(i)	M1 non-linear M2 bond angle – $120 > x \geq 116^\circ$	2
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